

Sprint 2 Review

1. Advisor Feedback – Areas to Improve from Sprint 1

- **Area Improve 1: Hardware Integration & Testing**
 - **Advisor comment:** Interfacing with other hardware components (such as the tracking system) remained untested during the Sprint 1 demo.
 - **Why this matters:** Hardware integration is a core dependency; until the software communicates with the spatial tracker, the system cannot function in a real-world environment.
 - **Sprint 2 change(s) made:** We successfully integrated the spatial tracker hardware and conducted testing using the laboratory machines to ensure data parity between the sensors and our software.
- **Area Improve 2: 3D Ultrasound Volume Reconstruction Implementation**
 - **Advisor comment:** The advisor anticipated that the 3D US volume reconstruction would be the most difficult and time-consuming technical hurdle of the project.
 - **Why this matters:** This is the primary deliverable of the software. Delays here would bottleneck the rest of the project milestones.
 - **Sprint 2 change(s) made:** We prioritized this module and have now mostly implemented the 3D volume reconstruction. This development phase allowed the team to move from theory to a functional prototype.
- **Area Improve 3: Mathematical Foundations (3D Transformation Chain)**
 - **Advisor comment:** The team had not yet fully demonstrated a deep understanding of the pivotal 3D transformation chains required for accurate reconstruction.
 - **Why this matters:** Accurate volume reconstruction relies entirely on the precision of the coordinate transformation pipeline (moving from image space to probe space to world space).
 - **Sprint 2 change(s) made:** Through the implementation of the reconstruction algorithm, we have successfully applied and demonstrated our understanding of the transformation pipeline, ensuring spatial accuracy in the rendered volumes.
- **Area Improve 4: Communication and Milestone Management**
 - **Advisor comment:** More frequent meetings were requested to ensure the heavy software implementation load met the established milestones.
 - **Why this matters:** Given the complexity of the project, consistent synchronization is required to prevent complex integration and ensure technical debt is managed.
 - **Sprint 2 change(s) made:** The team increased meeting frequency to stay aligned on rapid development cycles, which directly contributed to meeting our implementation goals for this sprint.

2. Peer Feedback on Sprint 1 Demo

Feedback in Forum

Peer Feedback Item 1: UI Complexity and Technical Language

- **Consideration status:** Implemented
- **Explanation:** Peers (William, Ashraf, Manav) felt the UI labels were too technical for a surgeon. We are refactoring the UI labels to use clinical terminology rather than math-heavy descriptions (e.g., "Registration" instead of "Matrix Transformation").
- **If implemented: related Sprint 2 change(s) or backlog item(s):** **S2-22: Data Capture** (Refine Start/Stop UI) and **S2-26: Volumetric Visualization** (GUI Integration).

Peer Feedback Item 2: Need for "Proactive" User Guidance (Pop-ups/Tooltips)

- **Consideration status:** Partially considered
- **Explanation:** Ashraf and Manav suggested adding pop-ups to explain why buttons are disabled. We have implemented "Progress Lights" to show completion, but the specific "help pop-ups" are currently in the backlog as we prioritize the core reconstruction engine.
- **If implemented: related Sprint 2 change(s) or backlog item(s):** **S2-12: Visual AR Validation Overlay** (includes user-facing status indicators).

Peer Feedback Item 4: Request for High-Level Architecture Visualization

- **Consideration status:** Implemented
- **Explanation:** Several peers (Amelia, Blake, Manav) requested a "quick drawing" or architecture diagram to see how the camera, sensors, and tools interact, noting that we spent too much time on raw code.
- **If implemented: related Sprint 2 change(s) or backlog item(s):** **High-Level Architecture Diagram** and **High-Level Object Design** added as primary documentation tasks for Sprint 2.

Peer Feedback Item 5: Demonstrating the Database "Use Case"

- **Consideration status:** Implemented
- **Explanation:** Mohammed suggested showing how a surgeon would actually use the stored session data. We have expanded our database schema to specifically support "Capture Session Data" that can be reloaded.
- **If implemented: related Sprint 2 change(s) or backlog item(s):** **S2-24: Capture Session Data** (Add new tables to database for session-specific retrieval).